CS430/910: Foundations of Data Analytics

Case Studies 2 | Dr Ayse Sunar

Objectives

- Going through the details of real-world applied data analytics project
- Understand the logic of the processes involved in data analytics
- See visualisation examples and their interpretations

Case Studies

- 4 papers on data analytics published in the scientific literature:
  - I Tube, You Tube, Everybody Tubes: Analyzing the World's Largest User Generated Content Video System
    Internet Measurement Conference 2007
  - What is Twitter, a Social Network or a News Media?
    19th international conference on World wide web 2010
  - Meme-tracking and the Dynamics of the News Cycle
    Knowledge Discovery and Data Mining (KDD), 2009
  - Link Prediction by De-anonymization: How We Won the Kaggle Social Network Challenge
    2011 International Joint Conference on Neural Networks (IJCNN)
- Read them!

Details on Case Studies

- Full details and links to the papers on course web site
  www2.warwick.ac.uk/fac/sci/dcs/teaching/material/cs910/
- Please read the papers in detail to get the full story
- Bias: papers are from Computer Science research community
  - Mostly address data analysis applied to large websites
  - The most well-studied example of “Big Data”
  - Examples should be familiar to you (YouTube, Facebook, Twitter)
- Objectives for the case studies:
  - To see examples of data analytics in practice
  - To introduce and motivate topics we will study in more detail later
  - To see examples of going from data to insight to understanding

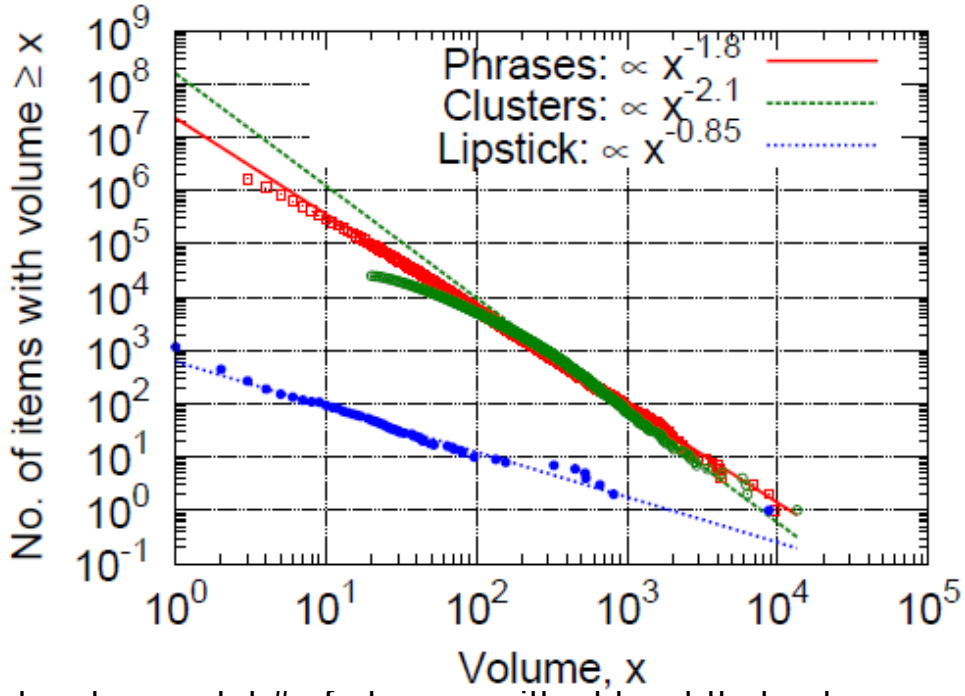
Case Study 3: Meme-tracking

- "Meme-tracking and the Dynamics of the News Cycle"
- Jure Leskovec, Lars Backstrom, Jon Kleinberg (Stanford, Cornell)
- International Conference on Knowledge Discovery and Data Mining (KDD), 2009
  snap.stanford.edu/class/cs224w-readings/leskovec09meme.pdf
- Objectives:
  - Track short distinctive phrases that travel through the web
  - Use this to study the "news cycle" in the news media

Data Collection and Preparation

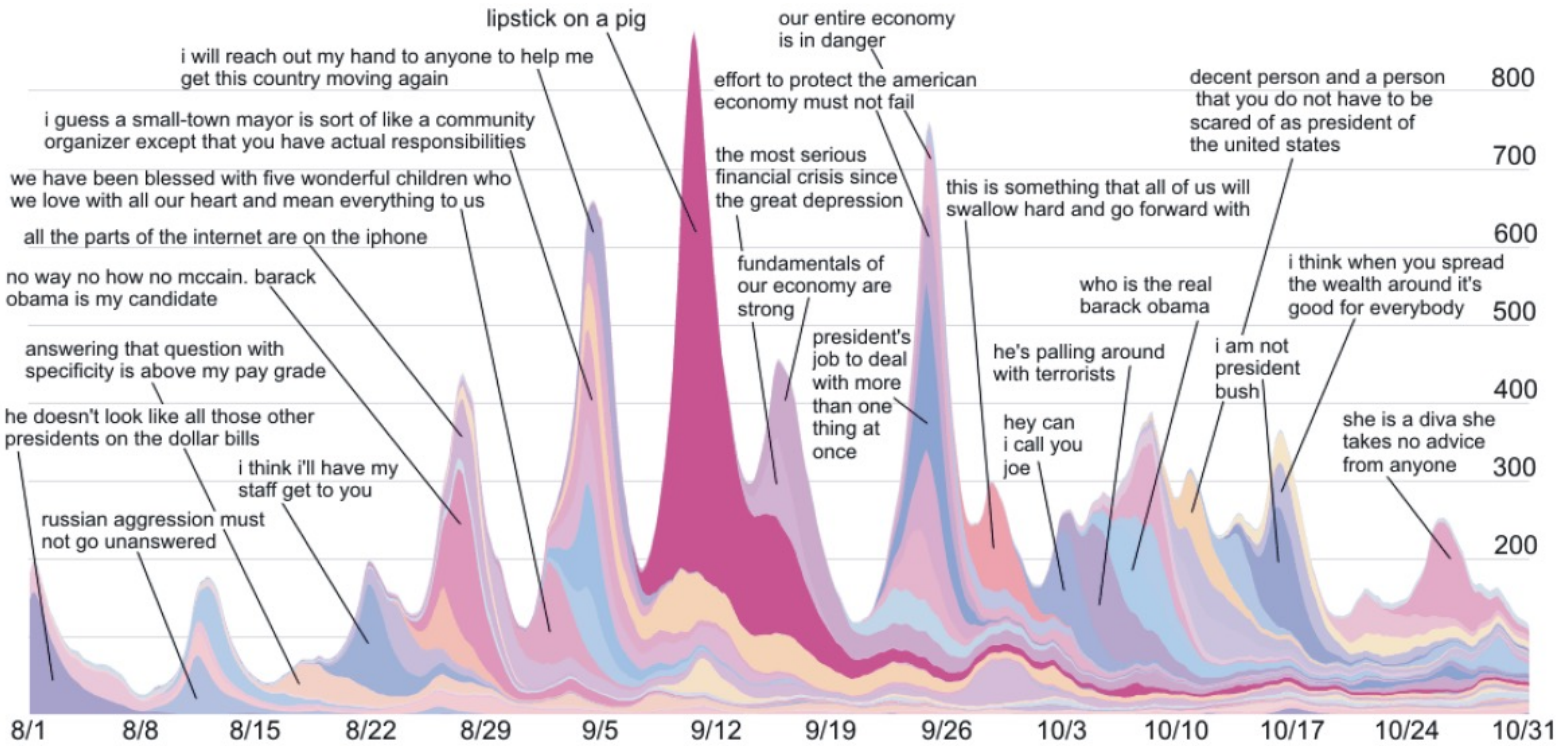
- News and blog activity from August 1 to October 31 2008
  - 90 million documents from 1.65 million sites
  - Used Spinn3r API to collect: see spinn3r.com
- Extracted 112 million phrases in “quotes”
  - Discard those with < 4 words or seen < 10 times (uninteresting)
  - Discard those where > 25% occurrences from same source (spam)
  - Leaves 47 million occurrences of phrases
- Collect phrases into clusters, based on overlap
  - Consider two phrases linked if they differ by at most 1 word
  - Partition the induced “phrase graph” to isolate key phrases
    - A long speech may have several key phrases in it
    - Quite detailed process; see paper for details

Phrase distribution



- For each volume, plot # of phrases with at least that volume
  - For all phrases, clusters of phrases, and phrases about “lipstick on a pig” (largest phrase cluster)

Most important threads

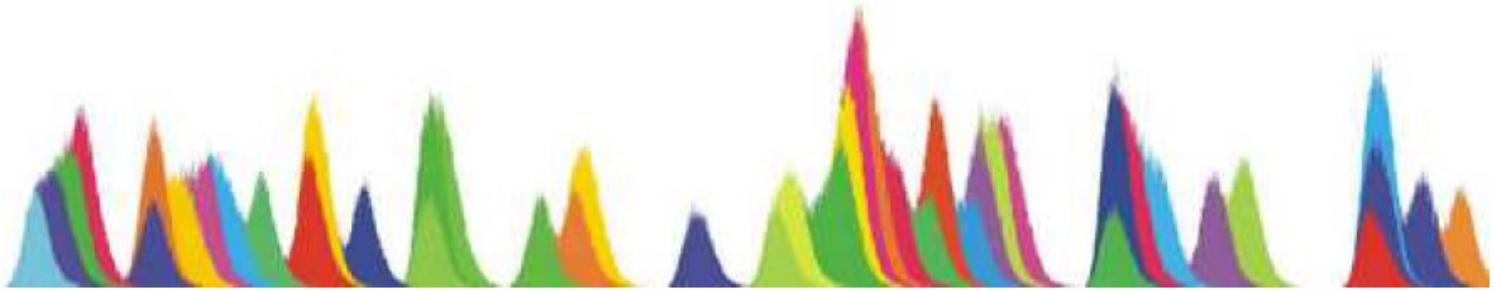


- Thread is all articles containing a phrase from a cluster
  - Plot is automatically generated and labeled to show volume

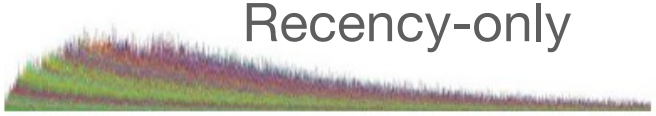
Formulating a model

- Advanced data analytics: propose a new model to explain data
- Try to capture major effects, and neglect minor points
  - Imitation: how likely one source is to replicate/copy/imitate content from another
  - Recency: news cycle dominated by recent events
- Model: simulate discrete time steps: news sources report threads
  - New thread produced at each step
  - At time t, each source picks thread j with probability $f(n_j)d(t-t_j)$
    - n_j: number of sources reporting on thread j ["Imitation"]
    - $d(t-t_j)$: decay factor based on age of thread j ["Recency"]
    - the likelihood of a meme being adopted based on factors the number of news sources mentioning it, temporal decay, and imitation probability, suggesting a dynamic spread of memes across sources and time

Validating the Model



meme propagation decaying rapidly over time, where older content is less likely to spread.



content spreading steadily as new sources imitate each other

- Simulation: pick f() as power-law and d() as exponential decay
  - Generates synthetic data that looks similar to real
  - Can we more rigourously validate the model?

Reflections on the paper

- A very innovative approach to the question
  - Created from scratch a way to think about “memes”
  - Proposed new models, and gave some evaluation of them
  - Tackled a timely question and used compelling examples
  - Plots and figures illustrate the key points
  - Widely cited (800+ citations)
  - But - models not robustly evaluated

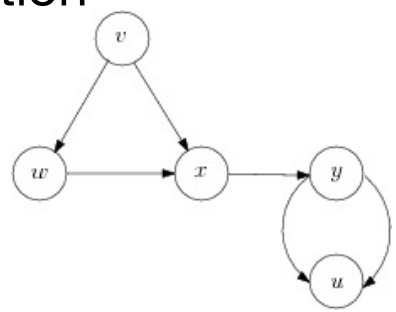
Case Study 4: Link prediction

- "Link Prediction by De-anonymization: How we won the Kaggle Social Network Challenge"
- Narayanan (Texas), Shi (Berkeley), Rubinstein (Microsoft)
- International Joint Conference on Neural Networks, 2011 http://arxiv.org/abs/1102.4374
- Objectives:
  - Correctly predict whether two users in a network would form a link
  - Use additional background information to improve the results

The Competition

Kaggle hosts competitions for data analytics

- Hosted the 2011 IJCNN Social Network Challenge in late 2010
- Provided a graph drawn from a social network
  - Nodes correspond to users in the network
  - (Directed) edges indicate a following relationship
  - Evaluation: determine whether a set of test edges (connection between users) truly occur
- It was later disclosed that the graph came from Flickr
  - Edges are (directed) “friendship” relations between users



Link Prediction

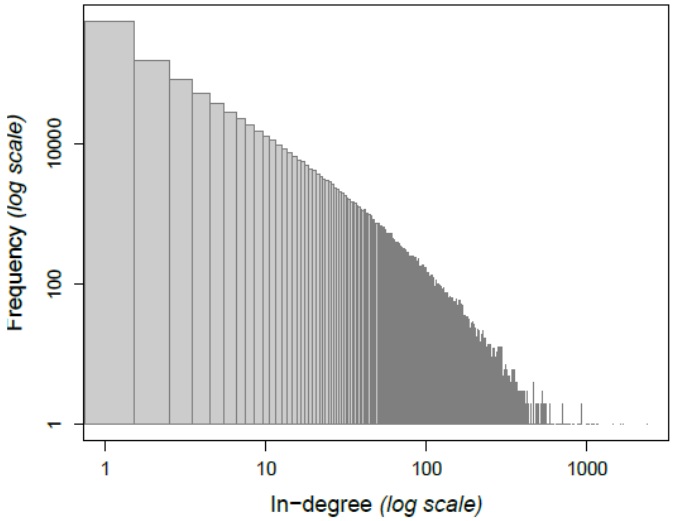
- Goal of “link prediction” is to determine which new links will form, given current state of the graph
- Many factors can be taken into account:
  - Properties of the nodes
  - Existing number of links
  - Common neighbours between a pair
  - Graph distance between the pair
- This work used an additional factor:
  - Try to match nodes in data set to their own data collection, i.e. integrate each node’s specific information (attributes, metadata, or interactions) within its own dataset

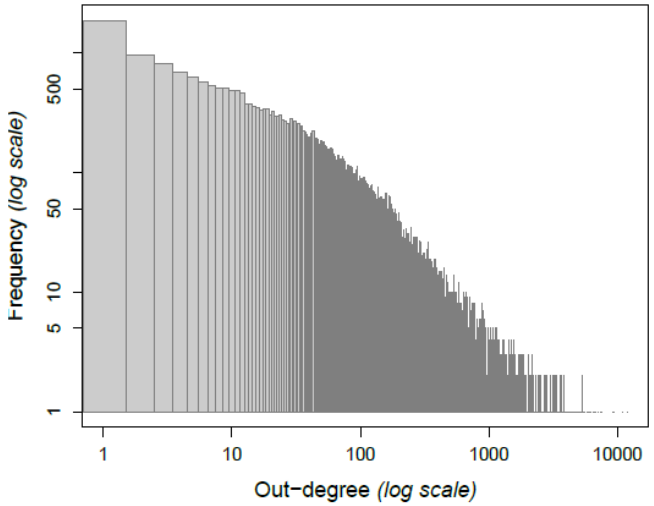
Data Collection

- Competition Data: Kaggle
  - 1.1M nodes, 7.2M edges provided as main data
  - 8960 “test edges”: 50% true edges (removed from main data)
  - 20% of test set held back by Kaggle to evaluate the results
- Competitors Data: Flickr
  - Crawled Flickr social graph (used Python + Curl library)
  - 2M nodes and all outgoing edges crawled
  - Total of 9.1M nodes, with 163M edges (much bigger data set)
- Evaluation: Area Under the Curve (AUC) [defined later]
  - Values are True/False, predicted as Positive/Negative
  - Ranges from 0.5 poor (random chance) to 1 (perfect)

Degree distribution

- Each node has an in-degree and an out-degree
- Skewed distribution, few nodes have high in/out-degree
  - Can try to use these as “landmarks”





Seed identification

- Try to find match a few nodes between the two graphs
  - Look at nodes with high in-degree (pointed to by many, many incoming edges)
  - These are likely to be present in both graphs
    - Because of the crawling process
- Pick highest n (20) degree nodes from Kaggle (K) and Flickr (F)
  - Try to match them up to get a “seed” matching
  - For a pair of nodes v, w in K (F), compute their “cosine similarity”:
    - #common neighbours(v,w)/√(#neighbours(v)*#neighbours(w))
  - Find best matching of nodes in K and F based on cosine similarities
    - Initially: manually
    - Later: optimization problem (see OR and optimization)

cosine similarity compares two nodes by assessing how many neighbours they share, normalised by the total number of neighbours each node has.

Propagation

- Now have matched n nodes between Kaggle and Flickr graphs
- Maintain a matching, and try to extend based on neighbors
  - Find pairs of nodes in Kaggle and Flickr whose similarity is high
  - Extend the matching. Iterate.
  - Some heuristics to accept a new pair into matching:
    - Must be at least 4 mapped common neighbors
    - Cosine-similarity score must be at least 0.5
    - Difference in similarity scores between best, and second best must be at least 0.2

Results

- Using “ground truth” information:
  - After 120,000 mappings in first stage, 99.3% correct
  - After second stage, had mappings for 14K out of 17.6K in test set
    - Overall accuracy for matched nodes 97.8%
    - A coverage of 57% of edges: still need to give answer for rest
    - For test edges, accuracy was 95%
- Use inferred information to predict links for more of test set
  - Look at all possible candidates for node pair in Flickr
  - Use these to vote on whether the edge is present or not
    - Accept if unanimous vote
  - Covers a further 19% of test edges

Machine Learning

- Leaves 24% of test edges without a mapping to Flickr
- Apply Machine Learning (the original goal of the challenge)
  - Create a number of "features" for each edge
    - In-degree and out-degree of node
    - Whether reverse edge exists
    - Measures of local graph (number of common neighbors etc.)
  - Train a "classifier" : see later lectures on classification
  - AUC for the classifier approach is ~0.9
- Total AUC for the whole approach on test data is 0.981
  - Excellent accuracy for deanonymized nodes, less for rest

AUC (Area Under the Curve) represents the probability that a randomly chosen positive example is ranked higher than a randomly chosen negative example by the classifier. An AUC of 1 indicates a perfect model, while an AUC of 0.5 suggests a model that is no better than random chance.

Reflections on the paper

- The score of 0.981 AUC was enough to win the competition
  - Second best was 0.969 www.kaggle.com/c/socialNetwork/leaderboard
  - The researchers contacted the organizers to reveal their method
  - Were told that this was within the rules
  - Read the messageboard for the competition to see other opinions www.kaggle.com/c/socialNetwork/forums
- Lesson: when understanding data, think beyond what you have
  - Are there other data sets that can help understand it better?
  - Can you learn properties of one data set and transfer to another?
  - Can you link two data sets to learn more about the first?
- Lesson: removing information does not “anonymise” data

Lessons from Case Studies

- Start by describing the data
  - How collected, with what attributes, what is dropped
- Look at the data
  - Plot combinations of attributes to see correlations/distributions
- Find models that agree with the data
  - Lines of best fit, long tail distributions
- Find explanations that are consistent with data and knowledge
- Extract observations of interest/value
- Make predictions and evaluate the quality

Acknowledgements

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Questions?